Who Wrote It? Identifying LLMs from Their Responses

CMPSC 448 midterm project.

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1 Problem & Dataset

1.1 Problem definition

Task. This is a 5-way text classification task. Each example is a triple **(LLM_name, LLM_Input, LLM_output)**, which we write as (y, p, o):

- $y \in \mathcal{Y} = \{\text{claude}, \text{gemini}, \text{llama}, \text{openai}, \text{qwen}\}$ is the model family (the class label),
- p is the user's prompt,
- o is the model's response.

We learn a classifier $f: x \mapsto \widehat{y} \in \mathcal{Y}$ for three choices of input x: the output only ($x = o$), the input only ($x = p$), or both ($x = (p, o)$). Model and company names in o are masked (Section 1.7). The classes are exactly balanced, so **chance accuracy is 20%**.

Evaluation. On a held-out test set of n rows, we report:

- **Accuracy**: $\frac{1}{n}\sum_i \mathbb{1}[\widehat{y}_i = y_i]$.
- **Macro-F1**: the unweighted mean of the five per-family F1 scores, $\frac{1}{5}\sum_{c \in \mathcal{Y}} F1_c$.

Neural results are the mean ± std over 3 seeds. Uncertainty is a 95% confidence interval from a **grouped bootstrap** that resamples test prompts. Two models are compared on the same test rows with the **exact McNemar test**, and an accuracy is compared with chance using an **exact binomial test** (details in Section 2.6).

Research questions.

- **RQ1:** can the family be identified from the output?
- **RQ2:** does the prompt help?
- **RQ3** (extra credit): do the fingerprints transfer across task domains?
- **RQ4** (extra credit): which characteristics distinguish the families?

1.2 Source and license

The data come from LMArena's public **arena-human-preference-140k** dataset on Hugging Face [2]. Each record is an anonymous "battle": one user prompt answered by two randomly paired models side by side, collected April–July 2025 on the Chatbot Arena platform [1]. The prompts are released under CC-BY-4.0; the

model outputs remain under each provider's terms of use. We did not query any model ourselves.

Personal information. We scanned all 9,665 prompts and outputs:

- **Prompts:** no e-mail addresses and no phone numbers. 10 prompts contain file paths with a username (e.g. `/home/<user>/`).
- **Outputs:** 65 contain an e-mail address, but most are placeholders such as `john@example.com`. Only 24 have a non-placeholder domain, and these are mostly public organizations or example servers. 13 contain a phone number, mostly public hotlines and business lines.
- **Repository:** the full dataset is not committed; it is regenerated from the public source. The committed 50-row sample excludes every row flagged by any detector.

1.3 Construction and filtering

All steps are in `scripts/prepare_arena.py` (seed 448), and each one is logged.

Step	Why	Result
Read all battles		135,634 battles
`language == "en"` and first battle of the session	English only; later battles in a session depend on earlier context	57,759 battles
Exactly one user turn and one assistant turn on both sides, text only (battles with images removed)	The text prompt must be everything the model saw	49,166 battles
Unpivot: one row per side	Each row is one (input, model, output)	98,332 rows
Map `model_version` to family by explicit prefix rules (e.g. `gpt*`, `o3*` → openai; `gemini*` but not `gemma*` → gemini); drop unmapped models	Family is the class label	78,762 rows
Drop empty prompts and outputs under 5 words	Almost no stylistic signal	78,282 rows
Keep the 5 largest families, sample 2,000 each; keep both sides of a battle together when they come from two different kept families	Balance, and the same prompt answered by two families, so the prompt cannot reveal the label	10,000 rows
Drop outputs that langdetect does not identify as English (code-only outputs kept)	The `language` tag describes the prompt; 318 outputs were in another language	9,682 rows
Downsample every family to the smallest count, keeping battle pairs first	Exact balance	**9,665 rows (1,933 per family)**

In the final data, 4,594 battles keep both sides, and **4,519 of the 4,978 distinct prompts are answered by two or more families**.

1.4 Schema

`dataset.csv` has the three required columns first, followed by four helper columns:

Column	Meaning
`LLM_name`	Family = class label (claude, gemini, llama, openai, qwen)
`LLM_Input`	The user's prompt
`LLM_output`	The model's response
`model_version`	Exact Arena model name, e.g. `claude-3-5-sonnet-20241022`
`prompt_id`	Short SHA-1 hash of `LLM_Input`; identical prompts share an id
`domain`	coding, math, creative_writing or general, from Arena's `is_code` and category tags
`source_id`	Arena battle id

1.5 Balance and composition

Family	Rows	Mean words	Median words	Coding	Creative writing	General	Math	Most common version (share)
claude	1,933	229.4	176	577	160	1,087	109	claude-opus-4-20250514 (18.4%)
gemini	1,933	760.6	682	591	175	1,036	131	gemini-2.5-flash (23.4%)
llama	1,933	471.8	449	600	152	1,033	148	llama-4-maverick-17b-128e-instruct (30.7%)
openai	1,933	426.4	367	623	158	1,028	124	chatgpt-4o-latest-20250326 (20.3%)
qwen	1,933	460.3	444	581	152	1,082	118	qwen3-235b-a22b-no-thinking (30.0%)

Each family mixes several model versions: 8 each for claude, gemini and openai, 6 for qwen and 4 for llama. No single version makes up more than 31% of its family.

1.6 Splits

We split **by prompt**, not by row: every row with the same `prompt_id` goes to the same split. This way a classifier is never tested on a prompt it saw in training. The split is 70/15/15, stratified by domain, with seed 448. That gives 3,484 / 747 / 747 prompts and 6,754 / 1,458 / 1,453 rows for train / val / test, with about 290 test rows per family.

As a check for near-duplicates, we normalized every prompt (lowercase, punctuation and whitespace removed) and re-hashed it. Only 11 normalized groups span more than one split, covering 10 test rows (0.7%). These are trivial prompts such as "how many r's are in strawberry", so we did not re-split.

1.7 Masking self-identification

Some outputs name their own model or company (e.g. “I’m Meta AI”). We replace whole-word, case-insensitive mentions of 16 model and company names (Claude, Anthropic, GPT, ChatGPT, OpenAI, o3, o4, Gemini, Bard, Google, DeepMind, Qwen, Alibaba, Tongyi, Llama, Meta) with `[MODEL]`. This changes 10.1–14.1% of outputs per family. **All experiments use the masked output** (`LLM_output_masked`) unless stated otherwise.

1.8 Known limitations

- **Time drift.** Which models were live on the Arena changed between April and July 2025, so family can correlate with when a prompt was asked.
- **Mixed versions.** Each family mixes several model versions, including “thinking” variants, so a family-level fingerprint is an average over them.
- **English only.** Only English prompts and outputs are included.
- **Prompt choice.** Arena prompts are whatever users chose to type: many are short, and some repeat (e.g. letter-counting questions).
- **Length differences.** Families differ strongly in output length: median 176 words for claude vs 682 for gemini (see Section 3.3).

2 Models & Training

2.1 Input representation

The tokenizer lowercases the text and splits it into words and single punctuation characters. Markdown symbols such as `#`, `*`, `-` and the backtick therefore stay as separate tokens, because formatting may be part of a family’s fingerprint. The vocabulary is built from the **training split only**, keeping tokens seen at least twice, plus `<pad>`, `<unk>` and `[SEP]`. This gives 57,838 entries. Sequences are cut to 512 tokens, keeping the beginning; 57% of training outputs are longer than that. In the input + output condition, the prompt gets at most 128 tokens, followed by `[SEP]` and then the output.

We compare three input conditions:

- a. **output only**: `LLM_output_masked`
- b. **input only**: `LLM_Input`
- c. **input + output**: `LLM_Input [SEP] LLM_output_masked`

2.2 TextCNN

Following Kim [3], tokens are embedded in 128 dimensions. One-dimensional convolutions with widths 3, 4 and 5 (100 filters each) slide over the sequence, so each filter detects a short n-gram pattern. After a ReLU, max-pooling over time keeps each filter’s strongest response anywhere in the text. The resulting 300 features pass through dropout (0.5) and a linear layer to the 5 classes. The model has 7.56M parameters, almost all of them in the embedding table.

2.3 BiLSTM

The BiLSTM uses the same 128-dimensional embeddings, followed by a 2-layer bidirectional LSTM [4] with 128 hidden units per direction and dropout 0.3 between the layers. Sequences are packed, so the LSTM never reads padding. The 256-dimensional outputs are **mean-pooled over real tokens only**, using the sequence lengths, then passed through dropout (0.3) and a linear layer. The model has 8.06M parameters.

2.4 Training

Both models are trained with:

- the AdamW optimizer [5] (learning rate 10^{-3}, weight decay 10^{-4})
- batch size 64 and cross-entropy loss
- gradient-norm clipping at 1.0
- early stopping on validation macro-F1 (patience 3, at most 20 epochs); the best checkpoint is restored before testing

Each configuration runs with seeds 448, 449 and 450, which fix Python, NumPy and PyTorch (with deterministic cuDNN). Training ran on a 4 GB Quadro M2000M GPU. On average the best epoch was 12.7 for TextCNN and 7.3 for BiLSTM (output only); one output-only run took about 3 minutes for TextCNN and about 6 minutes for BiLSTM.

The training curves (Figure 1) show the BiLSTM fitting the training set quickly: its training loss keeps falling while validation macro-F1 levels off after a few epochs, which is where early stopping helps.

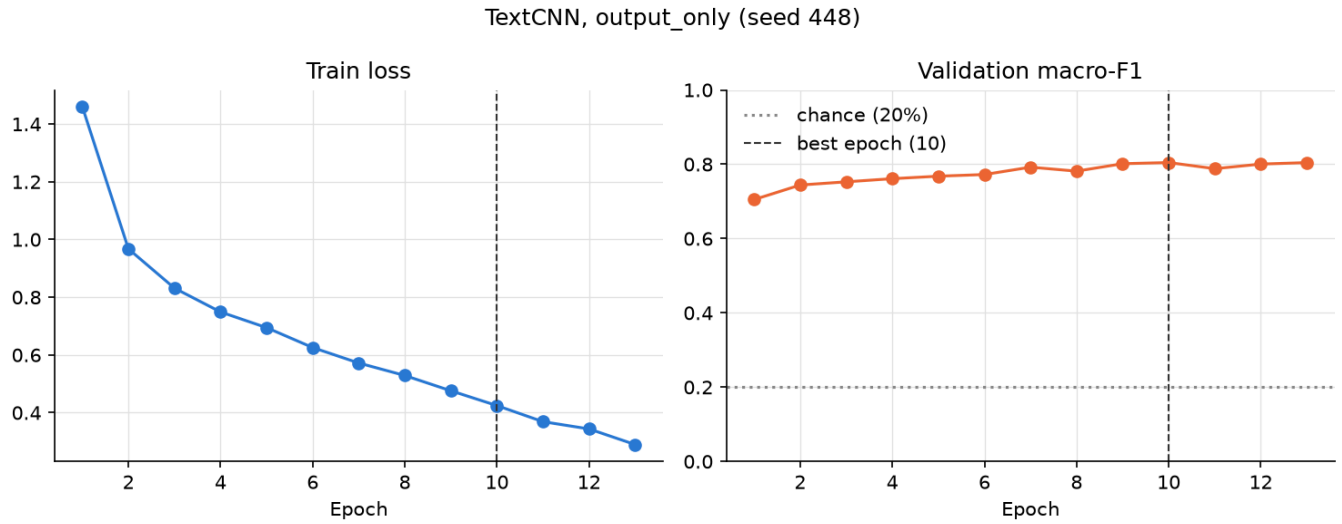


Figure 1a. TextCNN, output only (seed 448): training loss and validation macro-F1 per epoch.

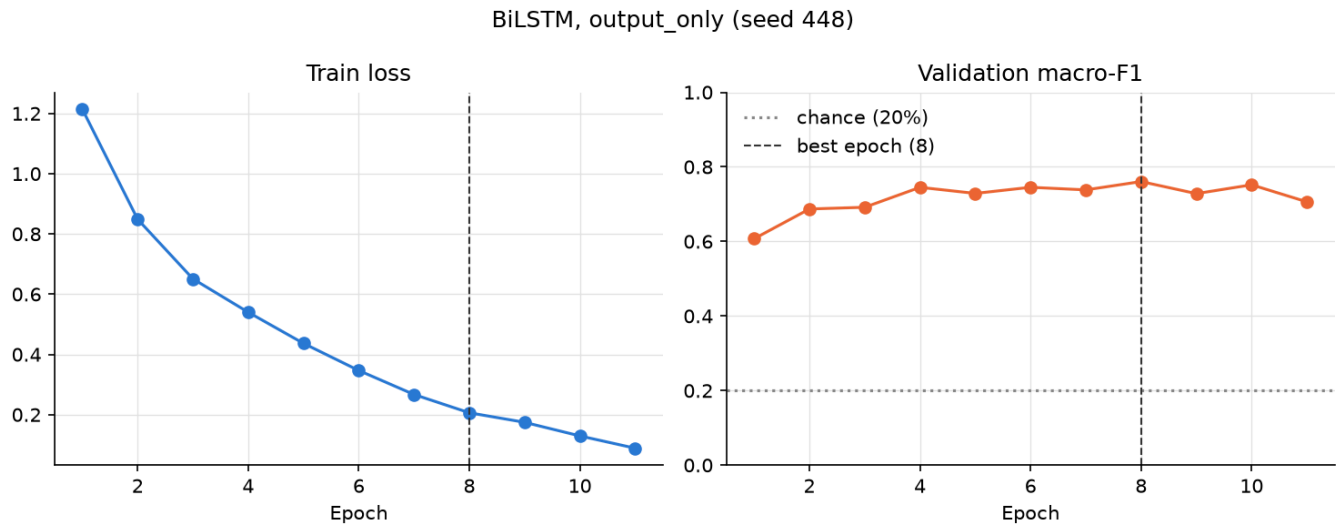


Figure 1b. BiLSTM, output only (seed 448): training loss and validation macro-F1 per epoch.

2.5 Baselines

1. **Majority class.**
2. **Length-only logistic regression** on four log-scaled lengths: characters, words, lines and newlines.
3. **TF-IDF + logistic regression** on word 1–2-grams plus character 3–5-grams (up to 100k features each, minimum document frequency 2, sublinear TF).

For both regression baselines, the regularization strength C is chosen on the validation set from {0.01, 0.1, 1, 10, 100} by macro-F1. The baselines are deterministic on fixed splits, so their three seeds give identical results.

2.6 Evaluation

We report test **accuracy** and **macro-F1**: the mean over the 3 seeds, the standard deviation over seeds, and a 95% confidence interval from a **grouped bootstrap** (2,000 resamples of test *prompts*, so the two rows that share a prompt are resampled together) [7].

To compare two models on the same test rows, we use the **exact McNemar test** [6] on seed 448. To compare an accuracy against chance, we use an **exact binomial test** against p = 0.20. Both tests treat rows as independent, but some test rows share a prompt, so their p-values are approximate. The grouped bootstrap does not have this problem.

3 Results

3.1 RQ1: can we identify which LLM generated a response?

Yes, well above chance. Table 1 shows the output-only results.

Method	Accuracy (%)	95% CI	Macro-F1	95% CI
Majority	20.0	[18.3, 21.8]	0.067	[0.062, 0.072]
Length-only LR	41.4	[38.7, 43.9]	0.399	[0.373, 0.424]
TF-IDF + LR	**86.3**	[84.4, 88.2]	**0.862**	[0.843, 0.881]
TextCNN	79.6 ± 0.5	[77.8, 81.5]	0.796 ± 0.004	[0.777, 0.814]
BiLSTM	75.6 ± 0.7	[73.6, 77.6]	0.757 ± 0.006	[0.736, 0.776]

Table 1. Output only, test set (1,453 rows). ± is the std over 3 seeds; CI from the grouped bootstrap. Chance = 20%.

Both required neural models are far above chance. TextCNN reaches 79.6% and BiLSTM 75.6%, and their confidence intervals do not overlap with each other or with chance.

The ranking is **TF-IDF + LR > TextCNN > BiLSTM**, and McNemar tests on the same test rows (seed 448) confirm both gaps:

Comparison	Only the first is right	Only the second is right	p-value
TF-IDF vs TextCNN	162 rows	59 rows	< 0.001
TextCNN vs BiLSTM	166 rows	111 rows	0.001

A linear model on sparse n-grams beats both neural models here. A likely reason is the data size: with about 6.8k training texts, an embedding table of 7.4M parameters (57,838 × 128) learned from scratch is harder to fit than regularized n-gram weights. Also, TF-IDF sees the whole text, while the neural models see only the first 512 tokens, and 57% of outputs are longer than that.

Errors are not spread evenly across families. With TF-IDF, per-family recall is:

Family	claude	gemini	llama	openai	qwen
Recall	94.2% (274/291)	90.7% (263/290)	91.2% (259/284)	79.5% (233/293)	76.3% (225/295)

Most remaining errors are **openai and qwen mistaken for each other**: 35 openai outputs were predicted as qwen, and 42 qwen outputs as openai (Figure 2). The neural confusion matrices show the same pattern (Figure 3).

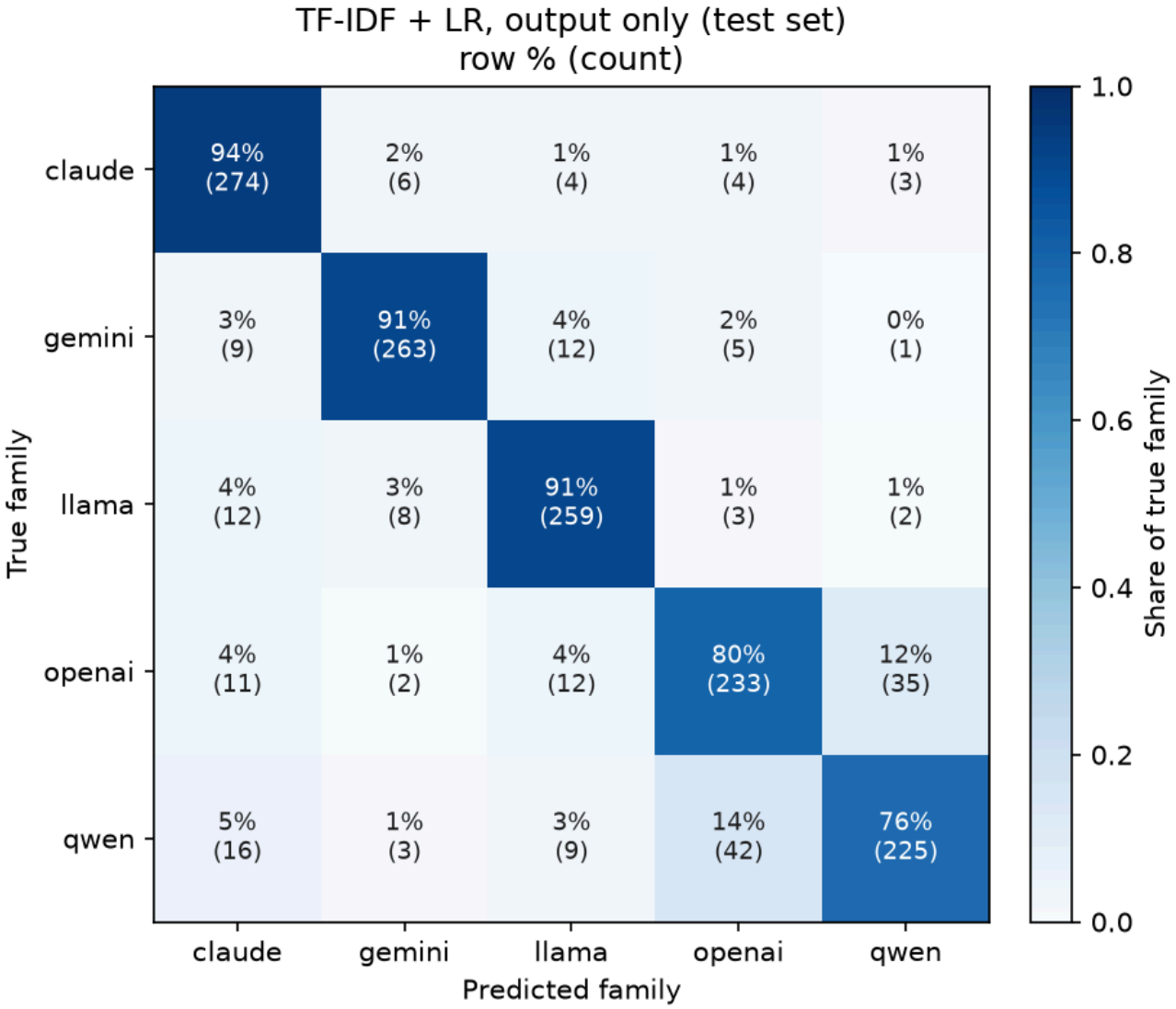


Figure 2. TF-IDF + LR confusion matrix, output only (test set), row-normalized.

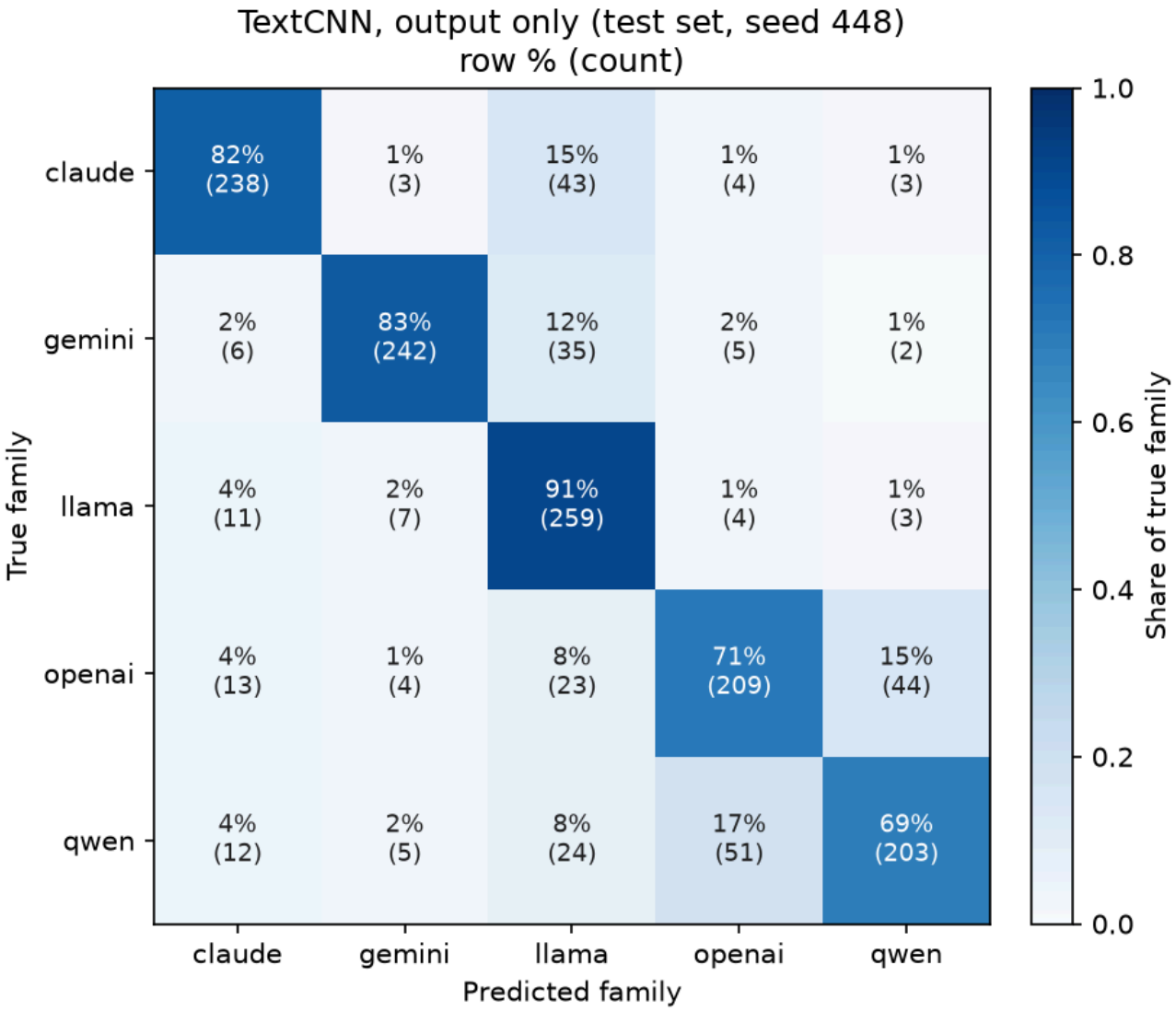


Figure 3a. TextCNN confusion matrix, output only (seed 448), row-normalized.

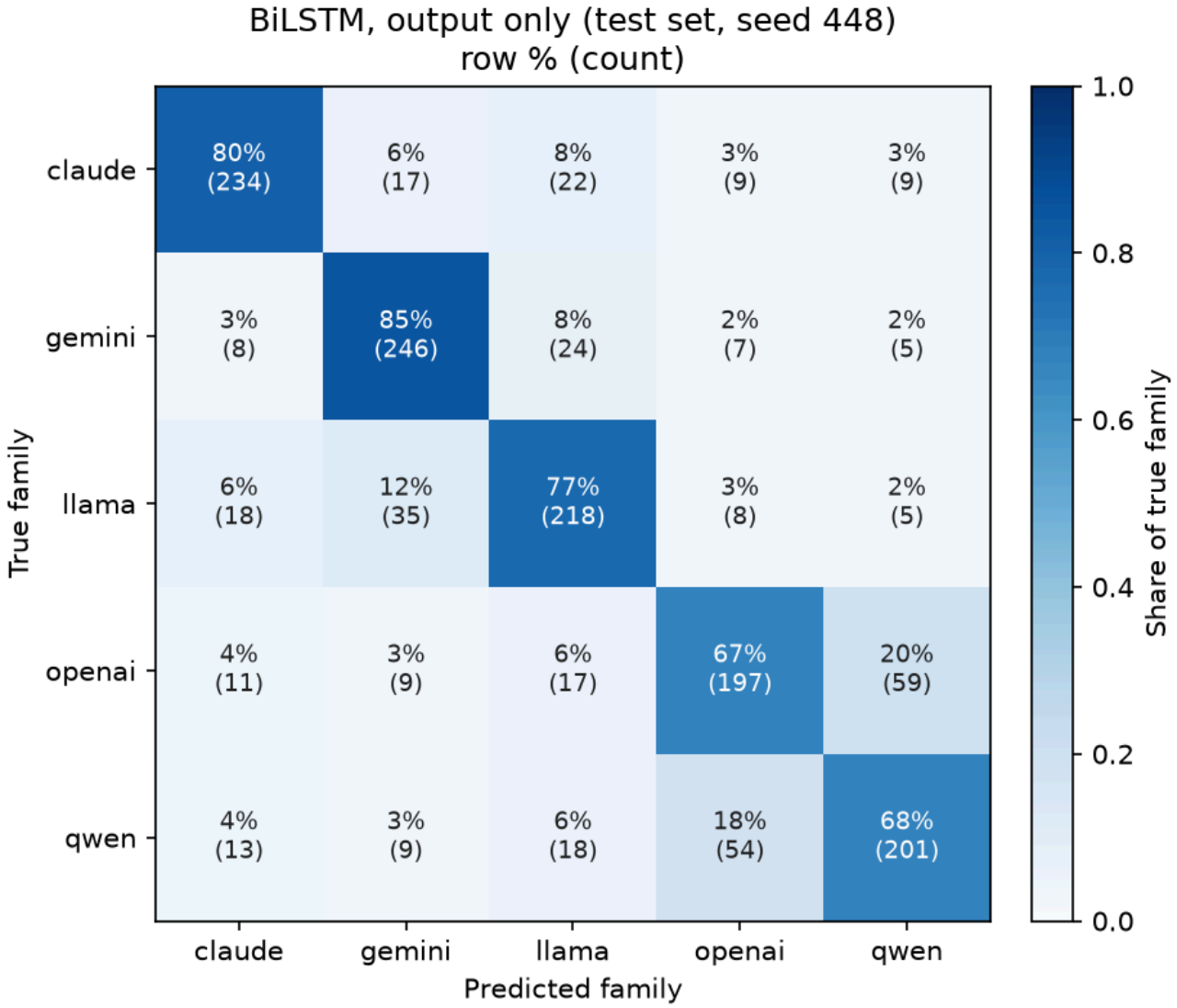


Figure 3b. BiLSTM confusion matrix, output only (seed 448), row-normalized.

3.2 RQ2: does the user's prompt help identify the LLM?

Method	Output only	Input only	Input + output
Majority	20.0 [18.3, 21.8]	20.0 [18.3, 21.8]	20.0 [18.3, 21.8]
Length-only LR	41.4 [38.7, 43.9]	20.7 [18.9, 22.4]	39.2 [36.7, 41.7]
TF-IDF + LR	86.3 [84.4, 88.2]	21.9 [20.1, 23.8]	85.1 [83.2, 87.1]
TextCNN	79.6 ± 0.5 [77.8, 81.5]	19.7 ± 2.0 [18.4, 21.0]	80.8 ± 0.3 [78.8, 82.8]
BiLSTM	75.6 ± 0.7 [73.6, 77.6]	20.6 ± 0.2 [19.2, 22.1]	75.0 ± 0.6 [73.0, 76.8]

Table 2. Test accuracy (%) for each input condition: mean ± seed std, [grouped-bootstrap 95% CI]. Macro-F1 CIs are in results/comparison_ci.csv.

(i) Can the family be predicted from the prompt alone? No. Every input-only model is at about chance (19.7–21.9%). Every input-only confidence interval includes 20% except TF-IDF's, whose interval [20.1, 23.8] only just excludes it, and its exact binomial test is not significant (21.9%, p = 0.077). In the binomial tests, no

model is significantly *above* chance. One TextCNN seed (448) is significantly *below* chance (17.8%, p = 0.039): its predictions are skewed toward some families. That is a bias, not information about the family, and it is 1 of 9 tests.

(ii) Does adding the prompt improve classification? We find no reliable improvement:

- **TF-IDF:** 85.1% with the prompt vs 86.3% without; the confidence intervals overlap almost entirely.
- **BiLSTM:** 75.0% vs 75.6%; McNemar p = 0.24.
- **TextCNN:** the only possible gain, 80.8% vs 79.6% (+1.2 points). This is larger than the seed std, but McNemar is not significant (88 vs 63 rows, p = 0.050) and the confidence intervals overlap. So any benefit is small and uncertain.

(iii) Why is input-only at chance here, and why might it work elsewhere? The cause is how the Arena collects its data: models are assigned to prompts largely at random, rather than users choosing a model for a task, so the prompt carries almost no information about which family answered it. In our data, this shows up as prompts that appear with answers from different families: because we kept both sides of a battle whenever possible, 4,519 of the 4,978 prompts were answered by two or more families. In datasets where each prompt is answered by only one model, or where users pick models by task (e.g. coding prompts sent to a coding model), input-only accuracy could be well above chance. It would then measure *who asks what*, not how a model writes. Time drift (Section 1.8) is the one route by which prompts could still carry a weak signal here. Our results show at most a very small one: TF-IDF's input-only accuracy is +1.9 points above chance and not significant in the binomial test.

3.3 Output length

Length alone carries real signal: the length-only baseline reaches 41.4%, about twice chance. But that is less than half of TF-IDF's 86.3%: measured above chance, length gets about +21 points and TF-IDF about +66. So length is far from the whole fingerprint.

Do the models only succeed on very short or very long outputs? Table 3 splits the test set into quartiles of output length. Accuracy rises from the shortest to the longest quartile for every text model, but the family mix also changes sharply across quartiles: 44% of the shortest quartile is claude, and 44% of the longest is gemini. To separate the two effects, we report two extra numbers per quartile:

- **Balanced accuracy:** the mean per-family recall, which the family mix cannot affect.
- **Accuracy expected from the mix:** the accuracy if each family kept its overall recall and only the family mix changed.

Quartile (words)	Share claude / gemini	TextCNN acc / balanced / from mix	BiLSTM acc / balanced / from mix	TF-IDF acc / balanced / from mix	Length-LR acc / balanced
Q1 (5–194)	44% / 9%	73.4 / 67.0 / 79.5	69.2 / 63.8 / 75.4	79.1 / 71.6 / 87.8	38.9 / 21.2
Q2 (195–395)	25% / 11%	74.9 / 75.3 / 78.7	73.4 / 74.0 / 74.7	83.6 / 83.0 / 85.9	39.0 / 34.9
Q3 (396–630)	6% / 17%	82.8 / 80.2 / 78.9	77.9 / 75.2 / 74.9	89.6 / 89.6 / 84.8	39.9 / 34.9
Q4 (633–5,705)	5% / 44%	87.5 / 83.6 / 81.5	82.1 / 77.8 / 77.7	93.1 / 89.5 / 86.8	47.8 / 27.2

Table 3. Test accuracy (%) per output-length quartile. Neural values are the mean over 3 seeds.

For the three text models, the family mix alone predicts an almost **flat** line (e.g. 79.5 → 81.5 for TextCNN). Yet balanced accuracy still rises by 14–18 points from Q1 to Q4. **The trend survives after accounting for the family mix:** longer outputs carry more evidence and are genuinely easier to attribute, and the models are not just succeeding on the extremes. Even the shortest quartile is well above chance: balanced accuracy is 64–72% for the text models.

The length-only baseline behaves differently. Its balanced accuracy is not monotonic (21% → 35% → 35% → 27%), so its higher raw accuracy in Q4 mostly reflects which families are long, not a general effect.

4 Analyses

4.1 RQ3: do LLM fingerprints generalize across domains?

Setup. We use leave-one-domain-out (LODO) evaluation on output-only text, with the same three models and hyperparameters as before. For each domain D (coding, math, creative writing, general), we train on the other three domains and test on D. The test set is always the rows of D in the original test split; training and validation rows come from the original train and validation splits, so the prompt grouping is unchanged.

For comparison, the **in-domain** condition trains on all four domains and tests on exactly the same rows of D. The only difference between the two conditions is whether D was seen in training.

To make the training sets equal in size, every one of them is subsampled to **593 rows per family (2,965 rows in total)**, the smallest per-family count over the four LODO folds. Which rows are drawn depends on the seed, and each condition runs with seeds 448–450. TF-IDF + LR therefore also gets a seed standard deviation here.

Because RQ3 trains on 593 rows per family instead of the full ~1,350, its absolute accuracies are lower than in RQ1, especially for the BiLSTM. Only the *difference* between the in-domain and LODO conditions is interpreted here.

Model	Held-out domain (test rows)	In-domain acc.	LODO acc.	Drop, acc. (pts)	Drop, macro-F1 (pts)
TF-IDF + LR	coding (444)	86.6 ± 0.9	83.7 ± 0.3	2.9 ± 1.3	3.0 ± 1.3
	math (99)	80.5 ± 2.1	75.8 ± 1.8	4.7 ± 3.5	5.0 ± 4.5
	creative writing (120)	75.6 ± 2.6	74.7 ± 1.3	0.8 ± 2.2 *(n.s.)*	1.2 ± 2.0 *(n.s.)*
	general (790)	82.5 ± 0.6	80.2 ± 0.2	2.4 ± 0.5	2.5 ± 0.5
TextCNN	coding (444)	77.9 ± 2.0	75.0 ± 0.4	2.9 ± 2.0	3.0 ± 1.6
	math (99)	72.4 ± 1.5	67.7 ± 0.0	4.7 ± 1.5	5.4 ± 2.0
	creative writing (120)	68.9 ± 2.9	63.9 ± 5.3	5.0 ± 3.3	5.5 ± 4.0
	general (790)	78.5 ± 0.7	76.8 ± 0.6	1.7 ± 1.1	1.8 ± 1.2
BiLSTM	coding (444)	68.0 ± 1.6	61.3 ± 4.3	6.8 ± 5.3	6.6 ± 5.4
	math (99)	71.4 ± 2.1	62.6 ± 2.7	8.8 ± 4.1	8.6 ± 2.6
	creative writing (120)	63.6 ± 2.1	60.3 ± 3.8	3.3 ± 1.7	4.1 ± 2.6
	general (790)	69.8 ± 1.0	66.4 ± 2.2	3.4 ± 3.1	3.6 ± 3.2

Table 4. RQ3, output only, size-controlled. Values are mean ± std over 3 seeds, in %. Drop = in-domain − LODO, paired by seed. n.s. = the drop is smaller than its std, so it is not meaningful.

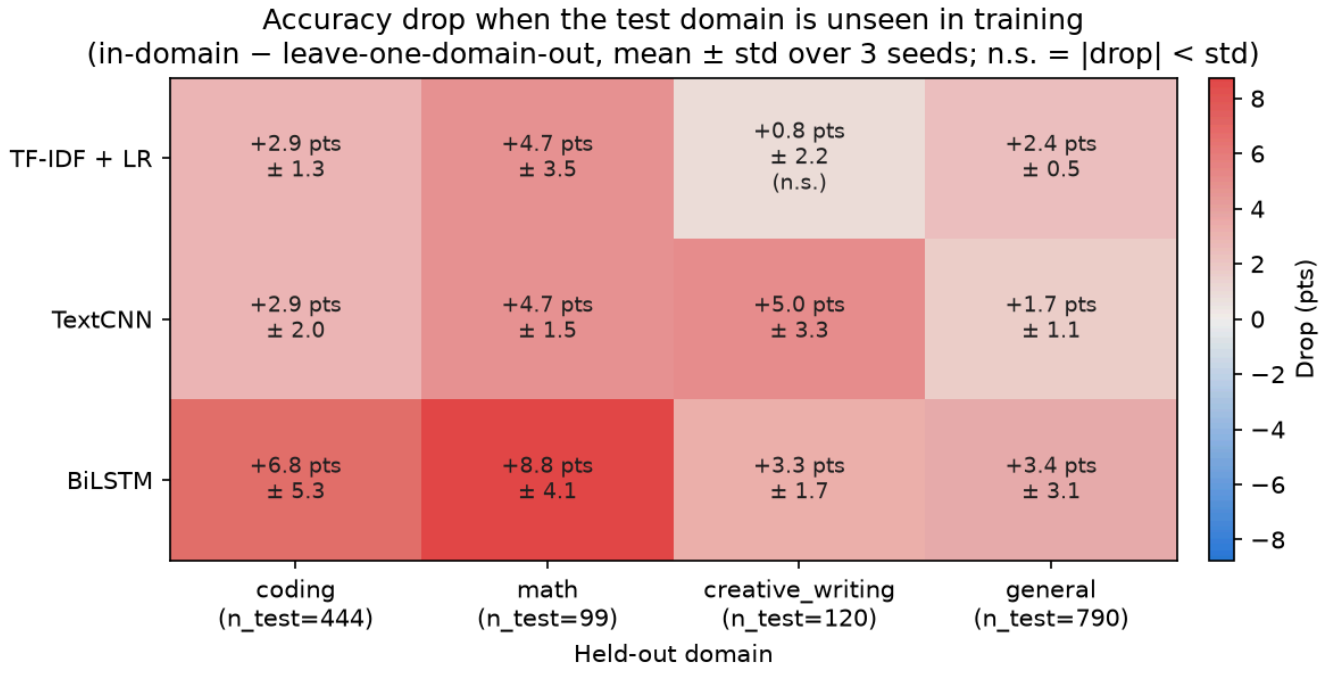


Figure 4. Accuracy drop when the test domain is unseen in training (in-domain – LODO), mean ± std over 3 seeds.

Findings.

- **Every domain is still identified well above chance when it was never seen in training.** The lowest LODO accuracy is 60.3% (BiLSTM, creative writing), against 20% chance.

- **The drops are a few points.** The average drop over the four domains is 2.7 points for TF-IDF, 3.6 for TextCNN and 5.6 for the BiLSTM. Measured above chance, the models keep 83%–98% of their in-domain accuracy, computed as (LODO − 20) / (in-domain − 20).
- **Only one drop is smaller than its std:** TF-IDF on creative writing (0.8 ± 2.2). Some others pass the rule only narrowly, such as BiLSTM coding (6.8 ± 5.3) and BiLSTM general (3.4 ± 3.1). With only 3 seeds, the std is itself uncertain, so these drops are probably real but their size is not precise.
- **Math is the hardest domain to *generalize to*.** It has the largest drop for TF-IDF (4.7) and the BiLSTM (8.8), and almost the largest for TextCNN (4.7, vs 5.0 for creative writing). It also has the fewest test rows (99), so these estimates are the least precise.
- **Creative writing is the hardest domain overall:** it has the lowest accuracy even in-domain (63.6–75.6%).

Per-family stability. Table 5 shows how much each family's recall drops when its domain is unseen. Cells marked † have fewer than 30 test rows for that family, so they are unreliable.

Family / model	Coding	Math †	Creative writing †	General
claude / TF-IDF	2.7	7.8 †	−1.2 †	−1.7
claude / TextCNN	0.4	11.8 †	4.8 †	4.2
claude / BiLSTM	0.0	−5.9 †	−7.1 †	0.2
gemini / TF-IDF	1.2	14.0 †	−2.6 †	4.2
gemini / TextCNN	−2.3	3.5 †	5.1 †	0.0
gemini / BiLSTM	11.6	22.8 †	19.2 †	14.7
llama / TF-IDF	3.7	−3.8 †	1.4 †	0.0
llama / TextCNN	18.3	−1.3 †	12.5 †	0.9
llama / BiLSTM	−4.5	10.3 †	4.2 †	−8.3
openai / TF-IDF	−1.0	10.0 †	−3.2 †	7.5
openai / TextCNN	9.6	13.3 †	−4.8 †	−3.2
openai / BiLSTM	9.6	11.7 †	1.6 †	7.1
qwen / TF-IDF	8.3	−2.0 †	11.1 †	1.8
qwen / TextCNN	−10.5	−2.0 †	6.4 †	6.3
qwen / BiLSTM	15.6	2.0 †	−1.6 †	3.0

*Table 5. Recall drop (in-domain − LODO, percentage points, mean over 3 seeds). Negative = the family is recognized **better** when its domain is unseen. † = low n (fewer than 30 test rows per family: math has 17–26 and creative writing 21–28; coding and general have 82–165).*

- **claude is the most consistently identifiable family.** On the two reliable domains (coding and general), its recall changes by at most 4.2 points for every model.
- **openai and qwen are the least stable.** Each has a drop of at least 7 points in a reliable domain for most models, for example openai 9.6 (coding) for both neural models and qwen 15.6 (coding) for the BiLSTM. These are also the two families most often confused with each other in RQ1.
- **gemini depends on the model.** It is stable for TF-IDF and TextCNN, but drops 11.6–14.7 points on the reliable domains for the BiLSTM.
- **Per-family numbers are noisy.** Their seed std is often 5–15 points (see the CSV), so only large, consistent differences should be read into this table.

Interpretation (hypothesis). The evidence points mostly to a **general fingerprint, with a smaller domain-specific part**: drops are small relative to the margin above chance, and they appear in every model. **Hypothesis (a):** the BiLSTM loses the most (5.6 points on average) because it learns more topic- or content-specific patterns, which do not carry over to a new domain. We did not test this directly. Its lower in-domain accuracy and higher variance across seeds could also explain part of the larger drops, so we do not claim it as proven.

4.2 RQ4: what characteristics distinguish the families?

Ablations: what happens when a signal is removed?

We retrain on transformed versions of the masked output; train, validation and test are always transformed the same way. The transformations are:

- **no_markdown:** removes headers, bold and italics, bullet and numbered-list markers, code fences, tables and horizontal rules.
- **no_opener:** drops the first sentence. This leaves 1.3% of training outputs empty.
- **length_matched:** cuts every output to 188 words, the dataset's 25th-percentile length.
- **no_markdown + length_matched:** both of the above.
- **unmasked:** uses the original output, with model and company names *not* masked.

For TF-IDF only, we also run **no_markdown + no emoji/punctuation** (emoji removed; dashes, curly quotes, ellipses, repeated punctuation and whitespace normalized) and **word_only** (no character n-grams). To see how much length signal survives truncation, we also run the length-only baseline on the length-matched text.

TextCNN runs 3 seeds. TF-IDF and length-only LR are deterministic, so they run once. Every drop is compared with the main masked result on the same test rows, with a paired, grouped-bootstrap 95% CI.

Variant	TF-IDF acc. [95% CI]	TF-IDF drop [95% CI]	TextCNN acc. [95% CI]	TextCNN drop [95% CI]
masked (reference)	86.3 [84.4, 88.2]	—	79.6 [77.8, 81.5]	—
no_markdown	84.2 [82.2, 86.2]	**2.1 [0.7, 3.5]**	75.0 [73.0, 76.9]	**4.7 [3.2, 6.2]**
no_opener	85.4 [83.5, 87.3]	0.9 [−0.3, 2.1] *(CI incl. 0)*	78.0 [75.9, 79.9]	**1.7 [0.5, 2.9]**
length_matched	83.4 [81.4, 85.3]	**2.9 [1.4, 4.4]**	79.0 [77.1, 80.9]	0.6 [−0.5, 1.8] *(CI incl. 0)*
no_markdown + length_matched	78.8 [76.6, 80.9]	**7.5 [5.6, 9.5]**	73.8 [71.8, 75.8]	**5.8 [4.2, 7.5]**
unmasked	86.4 [84.5, 88.3]	−0.1 [−0.6, 0.5] *(CI incl. 0)*	80.0 [78.1, 81.8]	−0.4 [−1.3, 0.6] *(CI incl. 0)*
no_markdown + no emoji/punct.	81.8 [79.7, 83.8]	**4.5 [2.9, 6.2]**	—	—
word_only (no char n-grams)	78.7 [76.5, 80.9]	**7.6 [5.7, 9.4]**	—	—

Table 6. RQ4 ablations, output only, test accuracy in %. Drop = masked – variant, in points; bold = the 95% CI excludes 0. The length-only baseline falls from 41.4 [38.7, 43.9] to 31.7 [29.4, 34.0] on length-matched text, a drop of 9.6 [6.8, 12.6].

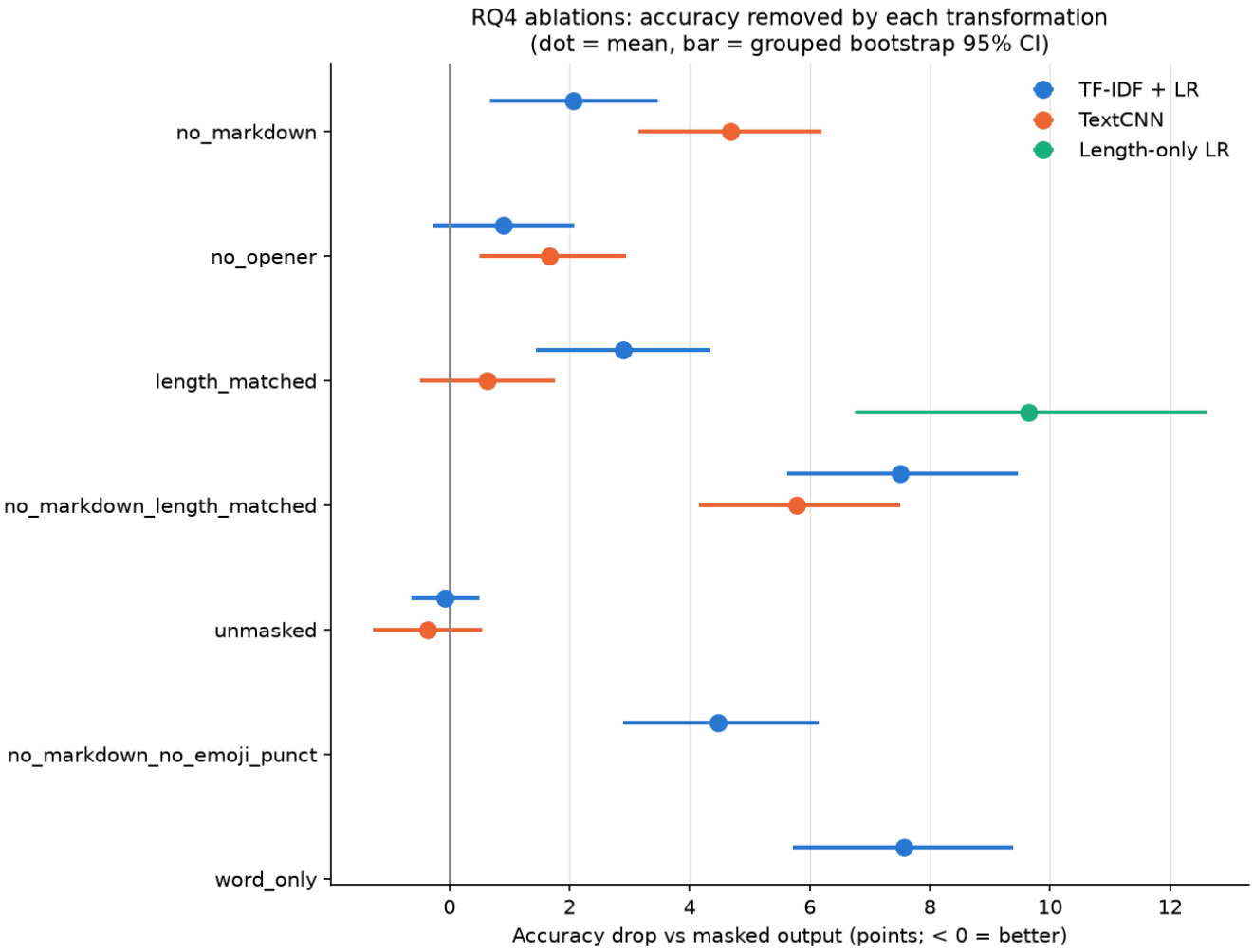


Figure 5. Accuracy removed by each transformation (dot = mean, bar = grouped-bootstrap 95% CI).

Descriptive features

Family	Median words	TTR (first 200 words)	Mean sentence length	Headers / 1k words	Bullets / 1k	Bold / 1k	Code blocks / 1k	Emoji / 1k (% of outputs with emoji)	Refusal rate
claude	176	0.607 (n = 785)	13.1	3.6	28.4	9.9	1.79	0.55 (1.6%)	6.2%
gemini	682	0.612 (n = 1,641)	12.3	2.7	32.0	32.0	0.87	0.22 (3.0%)	1.9%
llama	449	0.553 (n = 1,664)	12.4	2.6	30.6	29.4	1.20	0.29 (2.7%)	0.5%
openai	367	0.623 (n = 1,421)	10.0	10.0	33.5	26.0	1.54	4.68 (36.2%)	2.6%
qwen	444	0.612 (n = 1,573)	9.6	12.8	40.2	45.0	1.75	4.05 (36.1%)	3.6%

Table 7. Style features over all 9,665 masked outputs. TTR = type-token ratio (distinct words / 200) on the first 200 words, so length does not affect it; only outputs with at least 200 words get a TTR (n shown). Bullets include numbered items. Refusals = a refusal phrase such as "I can't" or "I'm sorry, but" in the first 200 characters.

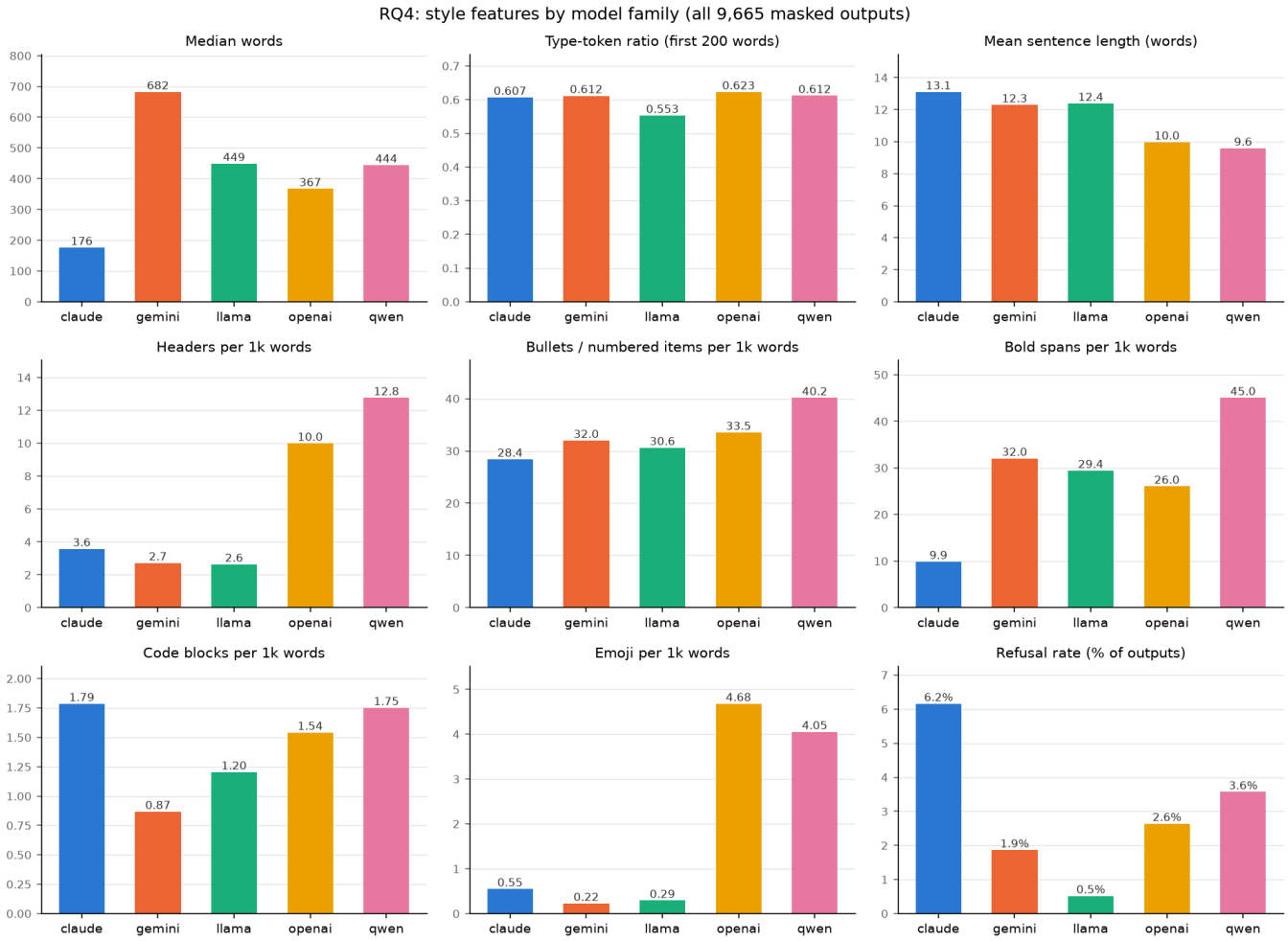


Figure 6. Style features by family.

Most distinctive words

These are the 15 word or bigram features with the largest positive weight for each family in the output-only TF-IDF + LR model. Features marked * are **prompt artifacts**: they come from repeated letter-counting prompts ("how many r's are in strawberry", answered "there are three…"), not from writing style.

Family	Top word / bigram features
claude	like me, me to, rs*, would, when, to explain, about, rather than, explain, several, most, they re, since, rather, through
gemini	okay, this is, let count*, okay here, are three*, is the, of course, very, within, course, core, in strawberry*, tell, tell me, yes
llama	here is, such as, ve, can be, such, and the, of the, however, are in, is, not, you re, the given, with that, be
openai	you want, so, want, if you, want can, summary, certainly, below is, etc, contains, sorry, sorry but, every, or, below
qwen	know if, me know, may, that request, let me, key, due to, sure ll, due, while, final, structured, tools, ensures, final answer

RQ4 summary: are the signals structural, lexical, or both?

Both, but mostly lexical.

- **Word choice carries most of the signal.** With markdown, emoji and punctuation all removed, TF-IDF still reaches 81.8%. With markdown removed and length matched, it reaches 78.8%, nearly twice the length-only baseline (41.4%).
- **The distinctive words are discourse habits**, not topic words: “okay here” and “of course” (gemini), “if you want”, “certainly” and “below is” (openai), “let me know”, “due to” and “final answer” (qwen), “here is” and “such as” (llama), “rather than” and “to explain” (claude).
- **The opening sentence is not essential.** Dropping it costs at most 1.7 points (TextCNN), and its CI includes 0 for TF-IDF. These habits appear throughout the text, not only in the opener (e.g. “let me know if” at the end).

Structural signals add a few points, and the families differ clearly on them.

- qwen uses 45.0 bold spans per 1k words, against 9.9 for claude.
- openai and qwen put emoji in about 36% of outputs, against 1.6–3.0% for the others.
- claude is the shortest (median 176 words) and refuses the most (6.2%).
- Removing markdown lowers accuracy for both models (CIs exclude 0).
- Removing emoji and normalizing punctuation costs TF-IDF another 2.4 points on top of removing markdown (4.5 vs 2.1).
- Type-token ratio is similar for four families (0.607–0.623) and lower for llama (0.553), so vocabulary richness separates at most one family.

Which single ablation hurts most?

- **Among the text transformations, removing markdown.** It is the clear largest for TextCNN (4.7 points).
- **For TF-IDF, removing markdown (2.1) and matching length (2.9) cannot be ranked:** their CIs overlap.
- **The largest single change overall is removing TF-IDF’s character n-grams** (7.6 points). Character n-grams capture punctuation, formatting symbols and sub-word patterns that the word features miss.

Three more observations:

- **Unmasked text is not classified better** (−0.1 and −0.4 points, both CIs include 0). Self-identification does not inflate the masked results.
- **Truncation does not remove all length signal.** The length-only baseline still reaches 31.7% on length-matched text: truncation can only shorten texts, so outputs that were already under 188 words keep their length.
- **Matching length has no clear effect on TextCNN** (0.6, CI includes 0). This is expected: TextCNN reads at most 512 tokens and max-pools over positions, which makes it largely insensitive to length by design.

Hypothesis (b): markdown and length are partly *overlapping* signals. For TF-IDF, removing both costs 7.5 points, more than the 5.0 that the two single drops add up to. This would fit a model that falls back on one signal when the other is removed. However, the CIs of the single and combined drops are wide, and for TextCNN the combined drop (5.8) is close to the sum (4.7 + 0.6), so we present this only as a hypothesis.

5 Lessons and Experience

5.1 Data collection was the hardest part of the project. Our first plan was to build our own dataset: 1,000 prompts sampled from Dolly-15k across 5 domains, sent to four models through OpenRouter. After 397 successful responses the account ran out of credit, and the free-tier alternatives had daily request caps that made the full set infeasible before the deadline. We switched to LMArena's arena-human-preference-140k. The lesson: estimate cost and rate limits before committing to a data plan, and check public datasets first. The switch also improved the design. Arena answers the same prompt with two randomly assigned models, which gave us the matched-prompt setup we had been trying to build by hand.

5.2 A well-designed dataset makes a null result informative. Input-only classification was at chance for every model (the best was TF-IDF at 21.9%, p = 0.077 against 20%). This is the expected outcome when models are assigned to prompts largely at random (Arena's pairing weights changed over time; see Section 1.8), and it is consistent with no leakage; the near-duplicate check (0.69% of test rows affected) is the direct evidence. Our first explanation, that prompts answered by two families cannot reveal either one, was incomplete: pairing is how the random assignment shows up in our data, not the cause. With unmatched prompts, such as a dataset where each model sees its own traffic, input-only accuracy would likely have looked "surprisingly good" for reasons that have nothing to do with fingerprints.

5.3 Shortcuts are everywhere, and baselines expose them. Median output length ranges from 176 words (Claude) to 682 (Gemini), and a length-only classifier reaches 41.4%, double chance. Between 10.1% and 14.1% of outputs name a model or company, so we masked those names. Unmasking did not measurably change accuracy (−0.1 points for TF-IDF, −0.4 for TextCNN; both CIs include 0), but we could only know that by testing it. Repeated prompts (for example, letter-counting questions) appeared among the top-weighted words, and near-duplicate prompts crossing splits affected 0.69% of test rows. Running majority, length-only and TF-IDF baselines before any neural model told us what "good" meant.

5.4 At this data scale, the simpler model won. TF-IDF with logistic regression (86.3%) beat TextCNN (79.6%), which beat the BiLSTM (75.6%), and both gaps were significant under McNemar's test. The BiLSTM's validation F1 plateaued by epoch 4–8 while its training loss kept falling. With about 6.8k training rows, sparse n-gram features are hard to beat, and extra model capacity mostly adds overfitting. One caveat: TF-IDF's regularization was tuned on validation, while the neural models used fixed hyperparameters.

5.5 Know which noise you are measuring. Our first significance rule compared differences against the standard deviation across 3 seeds. Under that rule, the +1.2-point gain from adding the input to TextCNN looked meaningful. Seed variation ignores test-sample noise, though: with 1,453 test rows, the standard error on accuracy is about 1 point, and McNemar's test did not confirm the gain ($p = 0.050$). We switched to bootstrap confidence intervals grouped by prompt, plus paired tests. Small subsets need the same care. Math has only 99 test rows, so per-family math results are indicative only.

5.6 An ablation is only as clean as its transformation. Our first "no markdown" ablation left emoji and punctuation in place, even though emoji appear in about a third of OpenAI and Qwen outputs. Our "length-matched" ablation truncated long outputs but left short ones untouched, so a length-only classifier still reached 31.7% afterwards. Similarly, our first accuracy-by-length analysis was confounded by which families dominate each length quartile, and only balanced accuracy separated the two effects. We learned to verify that a transformation removes what it claims to, for example by re-running a cheap baseline on the transformed data.

5.7 Engineering habits saved time. Training ran on a 4 GB laptop GPU (Quadro M2000M) in background jobs that stopped whenever the session ended, so we made every run resumable, fixed all seeds, used deterministic cuDNN, used only relative paths, and saved every result to a file instead of printing it. Every number in this report traces to a results file. Before publishing the code, we scanned the data for personal information and chose not to commit the full dataset. It is regenerated deterministically by our preparation script.

5.8 AI tools need verification. We used AI coding assistants to write and debug much of the code, which made a project of this scope feasible in the time available. Several interpretations they proposed were wrong or overstated, though, including the first input-only explanation, an early "mostly word choice" claim made before punctuation and emoji were controlled, and the input-helps-CNN gain. We treated their output as a draft and checked every claim against the saved results.

6 Food for Thought: one answer from outside the dataset

After the experiments, we gave the final model two answers that are not in the dataset. The model is TF-IDF + LR on masked outputs (output only), rebuilt from its saved settings. We checked that it reproduces the reported 86.3% test accuracy exactly, so it is the same model as in Section 3.

Reply (words)	True source	gemini	openai	llama	claude	qwen	Prediction
Stack vs queue, for a CS freshman (383)	not known	**57.8**	25.6	9.7	6.4	0.5	gemini
Why hydrogen is used as rocket fuel (639)	ChatGPT (openai)	**35.5**	28.7	21.3	11.5	2.9	gemini ✗

Table 8. Predicted probability per family (%). Both replies were pasted as plain text.

The ChatGPT answer was attributed to gemini. The code ran correctly; this is the kind of error the test results predict, made more likely by four measurable differences from the training data.

1. **The model was uncertain, not confidently wrong.** gemini led by only 6.8 points, and the true family came second. Even in-distribution, the model recognizes openai outputs only 79.5% of the time (233 of 293).
2. **The text no longer looks like the openai outputs the model learned from.** The reply has no headers, bold or emoji, against 10.0, 26.0 and 4.7 per 1,000 words for openai in training. Its sentence length (12.3 words) matches gemini (12.3) rather than openai (10.0), and its length (639 words) is close to gemini's median (682), not openai's (367). Only 25 of the 1,933 openai outputs in our data are at least 500 words long with no headers, bold or emoji. The most likely cause is that the answer's markdown was lost when it was copied as plain text, and formatting is part of the fingerprint (Section 4.2).
3. **The model has little evidence on this topic.** Only 31 training outputs mention rockets (11 of them from gemini), and none mention "specific impulse". The push toward gemini came equally from word and character features, and it mixed style words ("is the", "this is", "very") with topic words ("space", "exhaust", "engine") that say nothing about the author.
4. **ChatGPT has changed.** The model learned openai from 2025 versions (chatgpt-4o-latest, o3, o3-mini, gpt-4.1, o4-mini and smaller variants). The answer came from today's ChatGPT, whose style may have drifted: this is the time-drift limitation of Section 1.8, seen in practice.

What this suggests. A test accuracy of 86% describes text that resembles the test set. A single answer that is newer, unformatted and on an unfamiliar topic falls outside it, and the model then leans toward whichever family writes the most similar plain prose. An attribution tool used in practice should therefore report its top two families and its confidence, not one label, and should not be trusted on one plain-text answer. The stack-vs-queue reply shows the same pattern (plain text, predicted gemini), but its true source is unknown, so we do not count it either way.

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